Causal Inference for Intervention & Service Evaluations: A Practical Handbook

Jamie Wong

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Welcome

This is a practical handbook on causal inference methods for evaluating policy interventions and service changes within the NHS, with a focus on leveraging observational real-world data to generate robust evidence.

The handbook was written initially as part of an NHS England PhD Internship done jointly with the NHS North Central London Integrated Care Board and University College London, and will continue to be periodically updated with revisions following feedback from academics, data scientists, data analysts, clinicians, and public health practitioners.

Part I

Introduction

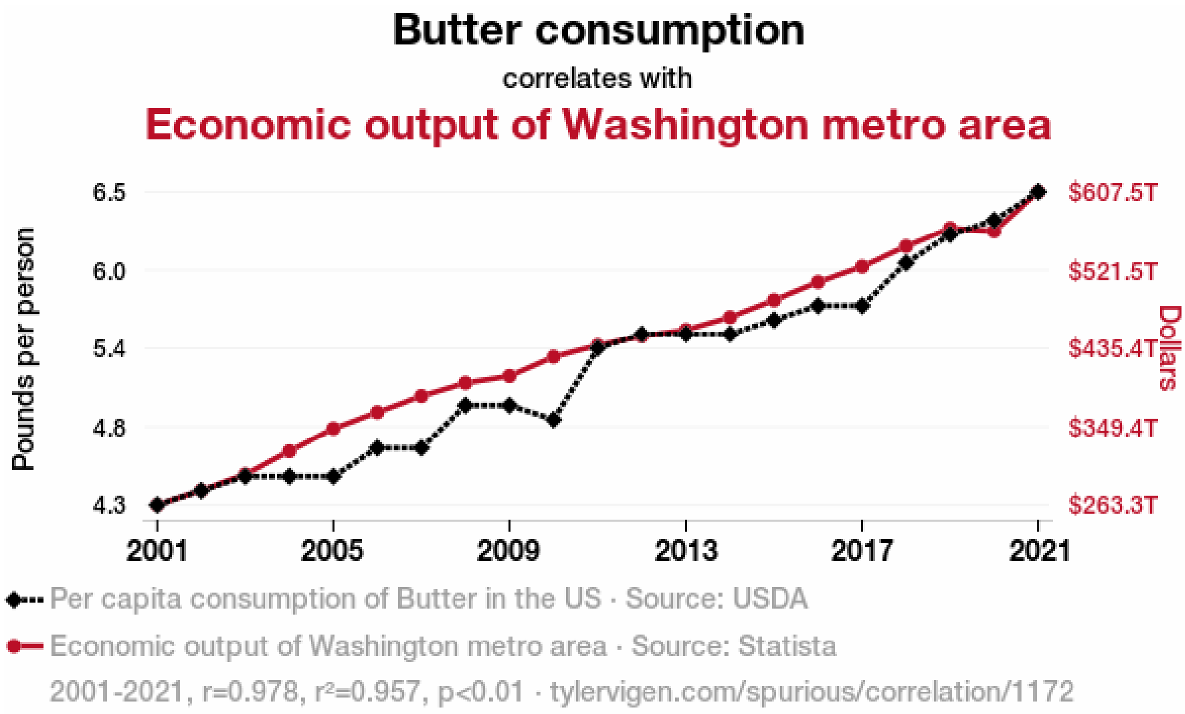
1 Background & Motivation for Causal Inference in Service Evaluations

When performing evaluations of healthcare services or interventions in the NHS, our key evaluation question of interest is almost always causal in nature:

- Has our new breast cancer screening programme reduced mortality?
- Did implementing a community-based model of care divert patients away from A&E services?
- Did implementing a hub and spoke model reduce clinical harm and lower patient waiting times?

Yet many of the methods currently used to evaluate interventions within the NHS are often descriptive in nature. Dashboards are frequently used in this manner, where any changes in an outcome of interest are fully attributed to a single change in implementation in services, without considering whether there are alternative explanations and/or other confounding factors are involved that could instead explain the change instead.

To better illustrate this point, Tyler Vigen publishes great examples of spurious correlations on his webpage (https://tylervigen.com/spurious-correlations), from which I've attached an example of below (Vigen 2024):



As you can see, butter consumption is positively correlated with the economic output of the Washington metro area. Needless to say, I highly doubt butter consumption was actually "greasing the wheels" of the local economy in the Washington metro area! It is therefore imperative that we don't conflate association with causation when we're conducting service evaluation as well.

Many books, papers, and reports have been published providing guidance on how one can estimate the causal effect of different interventions have on specific outcomes, in the absence of a randomised trial. Some resources have even been developed to help narrow down which types of methods would be most appropriate depending on the context. However, much of the guidance has been limited to methods originating from a specific field (e.g., econometrics, social sciences, epidemiology), rather than encompassing all approaches relevant to service and intervention evaluation within the realm of health and social care.

This handbook aims to bridge this gap in the literature, and establish new unifying guidance for selecting appropriate study designs and analytical methodologies for conducting causal inference in service evaluations. You'll find:

1. A decision tree diagram to aid with method selection
2. Concise overviews of each approach
3. Pointers to more in-depth resources for technical details about each method (including technical explanations in textbooks, applied examples in the literature, and practical coding exercises)

Do note that this handbook itself will not be going into too much technical detail about each analytical method, but instead serves as a resource to guide analysts in choosing the appropriate to use for their question.

I would also like to note that this handbook will primarily focus largely on established statistical methods for causal inference. More novel and experimental machine learning based methods do exist for causal inference. However, most of these methods require additional computational power, have components that are more opaque (akin to a black box), and many of these still haven't been extensively used in the literature yet. For the purposes of keeping this guide as approachable as possible, and our methods and assumptions as transparent as we can, this is the focus we have chosen for this handbook. An extensions section will be included at the end for those who may be interested in further delving into novel machine methods as well.

2 A Brief History on Conceptualising Causality

2.1 The Potential Outcomes Framework

The field of causal inference has a long history built on insights from multiple disciplines. Multiple schools of thought have been developed, contributing different ideas on how to conceptualise and define the concept of causality.

One of the most widely adopted ways to conceptualise causal inference in the fields of applied statistics, econometrics, and epidemiology is through the idea of potential outcomes, also referred to as the counterfactual approach to causality. This is where we aim to determine if the outcome for an individual would have been different if given different hypothetical value(s) of the exposure of interest.

Such counterfactual ideas stretch back to key texts written during the early to mid 19th century:

"If a person eats of a particular dish, and dies in consequence, that is, would not have died if he had not eaten of it, people would be apt to say that eating of that dish was the cause of his death" (Mill 1843).

However, it wasn't until Neyman's 1923 Master's thesis where the notation for potential outcomes was first formalised, introducing the idea that within randomized experiments, if we're interested in estimating the effect of a treatment or intervention, each unit in a study for a binary outcome can only have two fixed potential outcomes, one under treatment and one under control (Splawa-Neyman, Dabrowska, and Speed 1990).

In 1925, Fisher proposed randomizing treatments to study units/participants in order to derive unbiased inferences from experiments, though without reference to potential outcomes or estimating average treatment effects (Ronald A. Fisher 1925; R. A. Fisher and Yates 1990).

This changed in 1974, where Rubin formally established what we now refer to as the potential outcomes framework, extending these ideas to non-randomized observational studies, and writing formal notation for calculating the average casual effect that is used today (Rubin 1974). Holland later named this the Rubin Causal Model, and emphasised the fundamental problem of causal inference:

"It is impossible to observe [the value of the response that would be observed if the unit was exposure to treatment] and [the value that would be observed on the same unit if it were exposed

to the control], and therefore it is impossible to observe the effect of [treatment] on [the unit of interest]" (Holland, Glymour, and Granger 1985).

In short, if we're interested in evaluating a treatment, one cannot observe both hypothetical outcomes for a single unit or individual, one in a universe where the unit/individual of interest is treated, and another where the same unit/individual is not treated.

Because the potential outcomes framework mirrors the logic found in randomized trials, the school of thought has underpinned virtually every method covered in this handbook, including those with origins in econometrics and epidemiology.

2.2 Structural Causal Models and Directed Acyclic Graphs

However, alongside the counterfactual school of thought, competing theories were developed in parallel, the most prominent of which came from Pearl. Rather than framing causation in terms of counterfactual outcomes, Pearl popularised the development of Structural Causal Models (SCMs), which use Directed Acyclic Graphs (DAGs) and structural equations to present causal relationships and underlying assumptions in both a visual and mathematical way (Pearl 2009; Pearl and Mackenzie 2018).

I will be delving into the specifics of Directed Acyclic Graphs in their own section later into this handbook. As a brief overview though, DAGs are diagrams which include each of your variables (exposure, covariates, and outcomes) as nodes, connected by arrows that indicate a direct causal relationship between them. Digitale et al., from UCSF have a great tutorial on DAGs if you'd like a brief introduction on the topic (Digitale, Martin, and Glymour 2022).

Historically when these methods were introduced, the two schools of thought were completely split, with econometricians favouring the potential outcomes approach, while computer scientists favoured structural causal models. A great discussion on this is provided by Nobel laureate Imbens (an economist) more recently, who argues that although these two frameworks are complementary, the potential outcomes approach is often preferred for econometrics because it aligns naturally with study designs that exploit policy changes or other "as-if random" interventions, rather than collecting data on and adjusting for a wide range of individual variables mapped out on a DAG (Imbens 2020). This type of thinking was how the field of quasi-experimental methods originated, the specifics of which will be discussed in later sections of the handbook.

Overtime though, some fields such as Epidemiology and Public Health have found ways to combine both school of thought into a single workflow:

1. DAGs are drawn to visualise causal relationships, identify confounders of interest, present assumptions made transparently.

2. Use back door criteria on the DAG to select the minimal set of covariates that require adjustment of to make an unbiased estimate of the causal contrast of interest (e.g., average treatment effect, average treatment effect on the treated).
3. Apply appropriate statistical methods to adjust for the set of covariates, such as regression modelling, to estimate the target causal contrast, under the potential outcomes framework. This hybrid approach works particularly well in settings with clearly defined treatment contrasts (say, drug X vs. drug Y) and rich data from longitudinal studies and electronic health records, which are harder to come by in economics.

This hybrid approach works particularly well in settings with clearly defined treatment contrasts (say, drug X vs. drug Y) and rich data from longitudinal studies and electronic health records, which are harder to come by in economics.

If you are interested in reading further into the origins of the potential outcome framework and quasi-experimental designs, Cunningham's supplementary teaching material hosted on github as part of his "Causal Inference: The Mixtape" book goes into this in more depth (Cunningham 2025). His slides of the "Foundation of Causality" in particular focuses on the history of these foundational ideas. The introductory chapter of Morgan and Winships' "Counterfactuals and Causal Inference," as well as chapter 2 of Imbens and Rubins' "Causal Infernece for Statistics, Social, and Biomedical Sciences" also provide a great overview of the historical context behind how causal inference as we know it today was first formalised (Morgan and Winship 2014; Imbens and Rubin 2015).

2.3 Additional Approaches to Establishing Causality

During the mid-19th century, before Rubin had established the potential outcomes framework, Hill had notably published a seminal paper covering a set of criteria that would help establish causality within epidemiological studies, now commonly referred to as the "Bradford Hill Criteria." These included: strength, consistency, specificity, temporality, biological gradient, biological plausibility, coherence, experimental evidence, and analogy (Hill 1965). The criteria, however, did not provide a practical way for epidemiologists and social scientists more generally to design observational studies targeting causal inference though. The criteria is therefore used more so as a guide for evaluating whether associations are causal while evaluating a set of evidence, and still often used in policy evaluations. Chapter 2 of Lash et al.'s textbook "Modern Epidemiology" provides a great explanation of each criteria if you'd like to delve deeper into this.

Additional causal frameworks have been built on top of this foundation as well, namely the Target Trial Emulation framework by Hernan and Robins which will be discussed in further detail in its own section (Hernán and Robins 2016). The framework aims to overcome many of the methodological issues facing classical epidemiological analyses by framing causal questions of interest within an ideal pragmatic randomized trial. This will underpin much of the methods covered as part of the "Epidemiology Methods" section further into the handbook.

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30 About the Author & Acknowledgements

30.1 About Me

Hi, my name is Jamie and I'm an MRC funded PhD student at UCL, focusing on the application of causal inference methods within Epidemiology, Data Science, and Health Economics.

Before my PhD, I previously graduated with a BSc in Population Health Sciences (Data Science) from UCL, and an NIHR funded MSc in Epidemiology from the London School of Hygiene and Tropical Medicine.

Prior to my PhD, my interest in researching the application of causal inference methods in observational data stemmed from my varied experience working across multiple areas of public health research, including: • Performing secondary analysis on a nutritional labelling RCT at the NIHR Obesity Policy Research Unit • Supporting cancer trials as a Clinical Trials Assistant in the MRC Clinical Trials Unit • Identifying barriers to STI testing at the NIHR Health Protection Research Unit in Blood Borne and Sexually Transmitted Infections • Quantifying the effects of acute and long-term exposure of air pollutants on respiratory related hospital admissions and related outcomes at UCL • Evaluating the risk of long-term adverse events in cancer survivors using electronic health record data at LSHTM

Having worked with both randomized trials and observational data, focusing my research on causal inference in real-world data felt like a natural next step.

If you have any questions do feel free to reach out to me via the following channels: UCL Email: jamie.wong [at] ucl.ac.uk NHS Email: laichunjamie.wong [at] nhs.net LinkedIn: https://www.linkedin.com/in/jamiewlc/ GitHub: https://github.com/jamiewlc

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